

Robotics Workshop · Fall 2026

Machine Learning for Fault Detection



UNIVERSITY OF
WATERLOO

Pearl Sullivan Engineering

IDEAs Clinic
10 Years



TODAY'S ROADMAP

01



Why Fault Detection?

Motivation & engineering examples

02



Machine Learning Basics

From data to learning

03



Classical Machine Learning

Core algorithms & hands-on practice

04



Deep Learning

Neural networks for complex data

05



Anomaly Detection

Finding unusual behavior

06



Edge Impulse

Practical setup & deployment

WORKSHOP RESOURCES

Everything you'll need today,
in one place.

COLAB NOTEBOOKS • CODE • WORKSHOP FILES

SCAN



OPEN GITHUB



FOLLOW ALONG



[GITHUB URL Link](#)

SCAN OR Click TO OPEN



Why Do We Need Fault Detection?



Detect early. Act before failure.



Safety

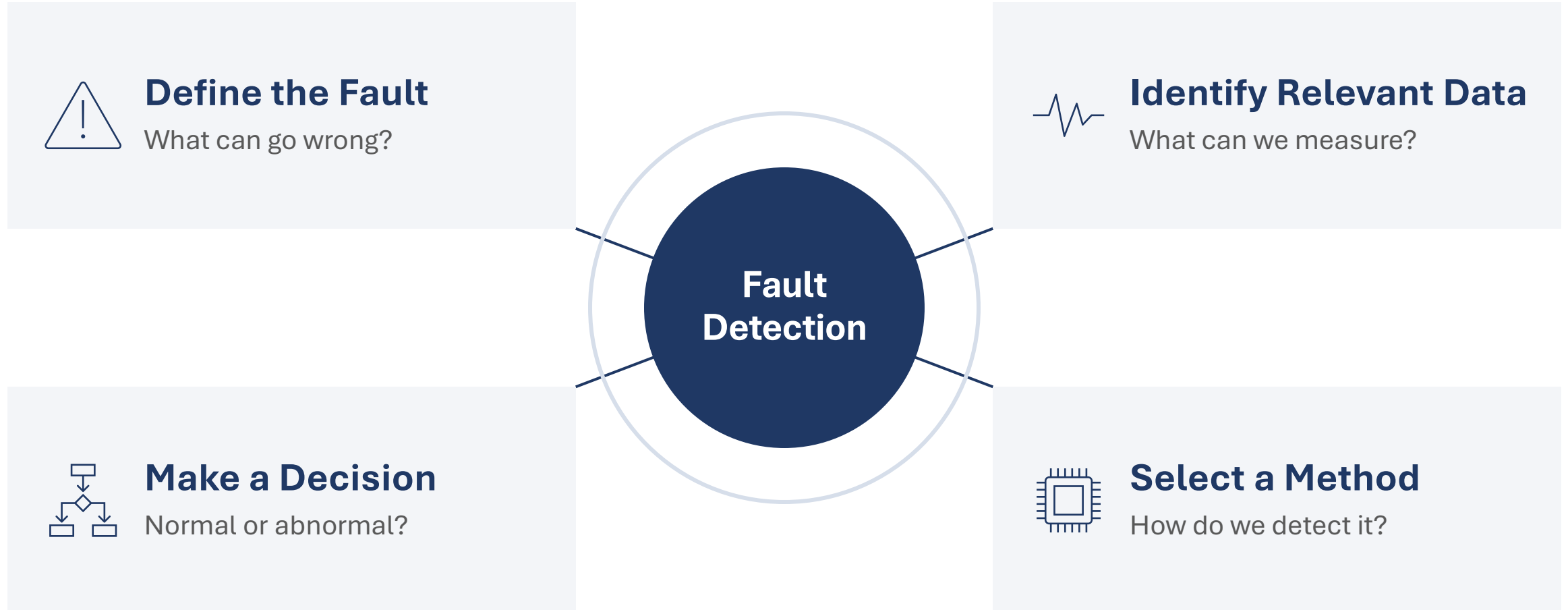


Downtime



Cost

How Do We Detect a Fault?



ENGINEERING EXAMPLES

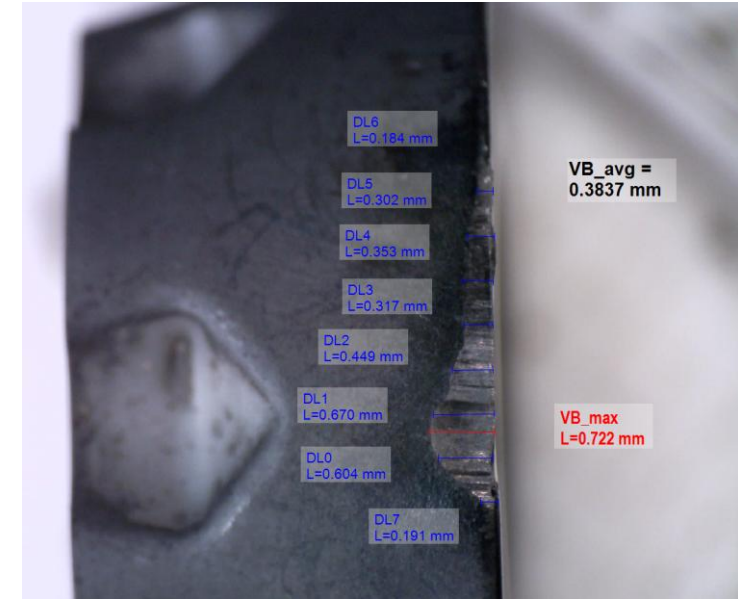
Fault Detection in Practice

Can you think of a system where detecting a fault early would matter?

Tool Wear Detection for CNC Machines

CNC (computer numerical control) machines cut metal and other materials automatically, following programmed tool paths.

- **Failure Type:** Flank Wear
- **Data:** Cutting Forces & Vibrations
- **Algorithm:** PINN + Transformer



Data Collection Setup

Data Collected



MOTOR CURRENT

AC current



VIBRATION

Tri-axial · X · Y · Z



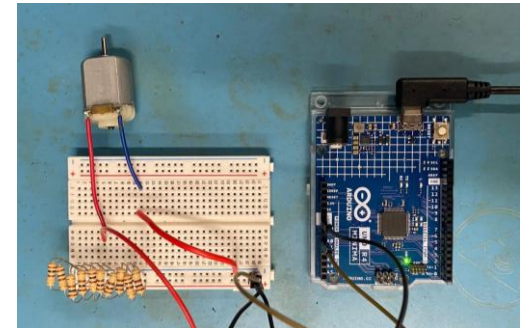
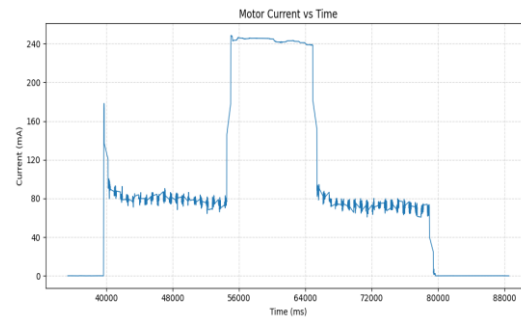
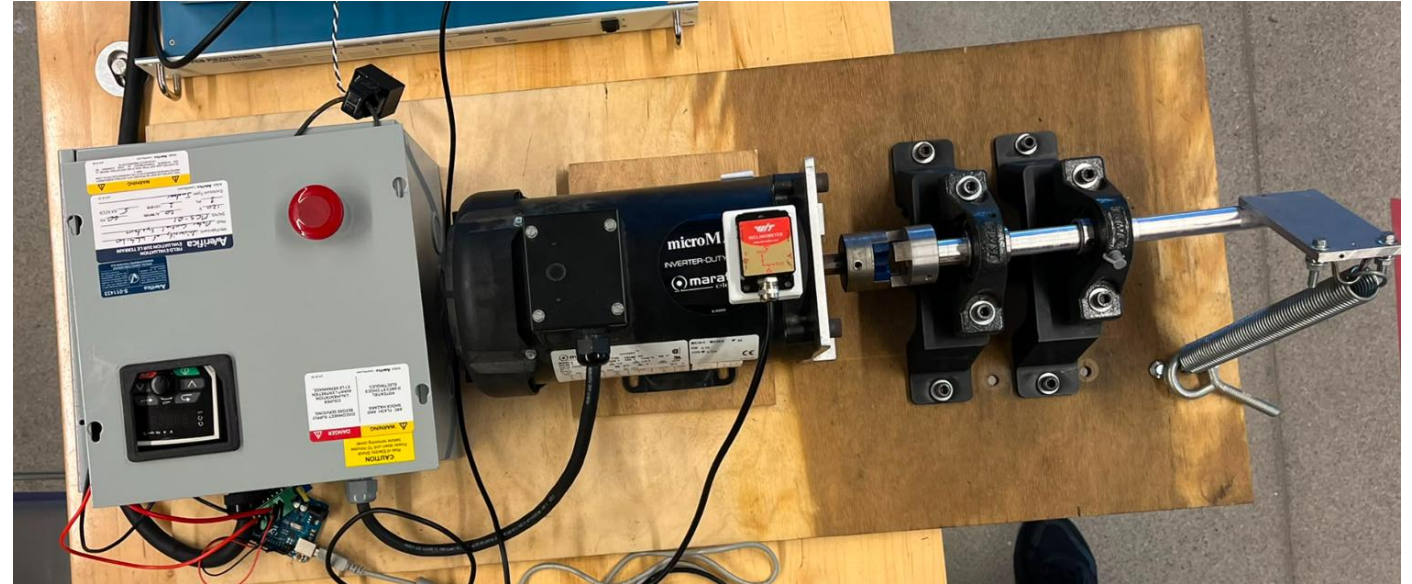
ORIENTATION

3-axis · Pitch · Roll · Yaw



ANGULAR VELOCITY

3-axis · X · Y · Z



MACHINE LEARNING

From Programming to Learning from Data

01

**Traditional Programming
vs. Machine Learning**

02

Machine Learning within AI

03

**When to Use Machine
Learning**

Traditional Programming vs. Machine Learning

Where does each one go?

INPUT

RULES

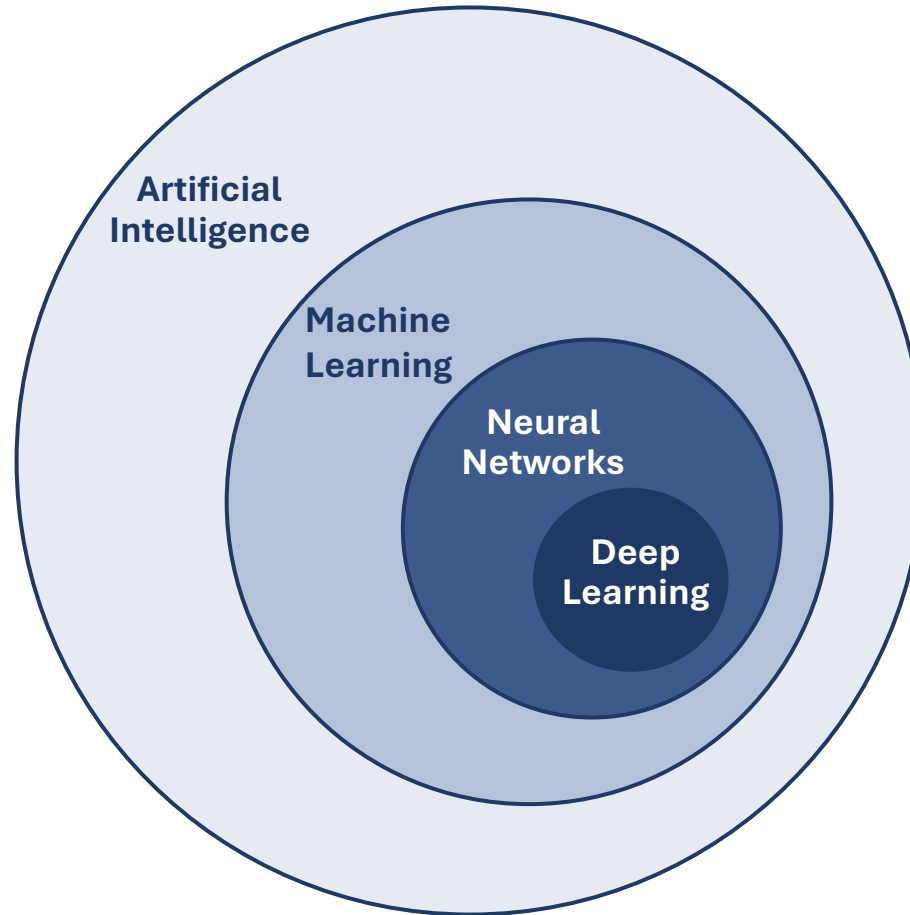
OUTPUT

**Traditional
Programming**

Machine Learning

Learned Model (Rules)

Where Does Machine Learning Fit?



Machine Learning is one approach within the broader field of AI.

When Should We Use Machine Learning?

Do we know how to explicitly write the rules?

Yes

Traditional Programming

- Rules are known
- Relationships are well defined
- Behaviour can be explicitly programmed

No

Machine Learning

- Rules are difficult to define
- Patterns are hidden or complex
- Data/examples are available

TYPES OF MACHINE LEARNING

How Can a Machine Learn?

01

**Supervised
Learning**

02

**Unsupervised
Learning**

03

Semi-Supervised
Learning

04

Reinforcement
Learning

Types of Machine Learning

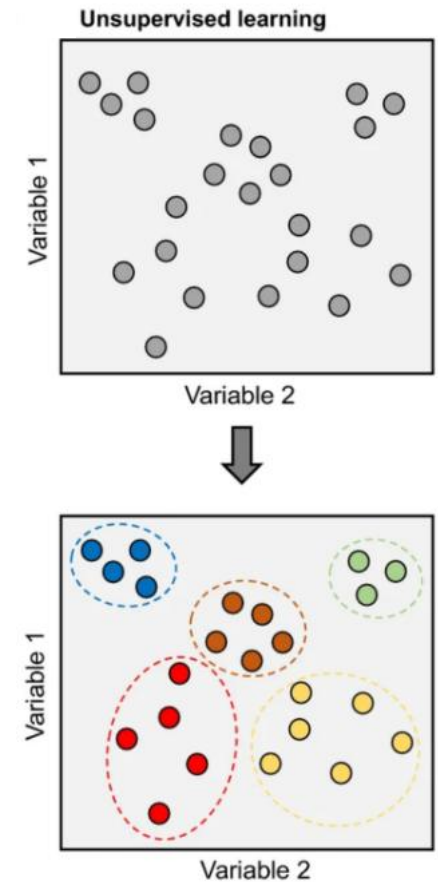
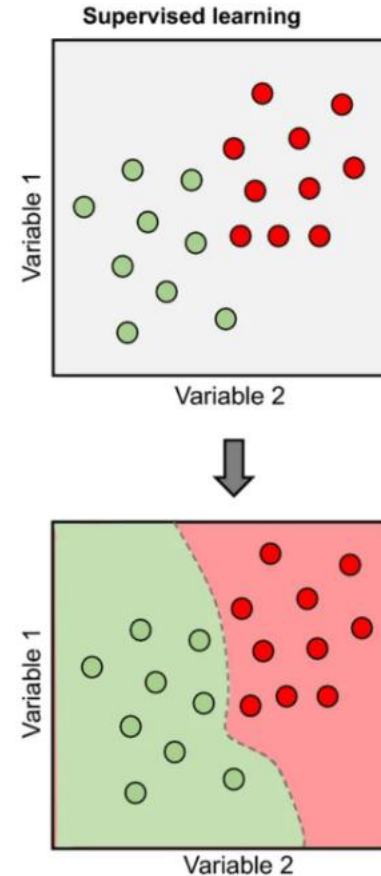
Supervised Learning

Learn from labeled data

Unsupervised Learning

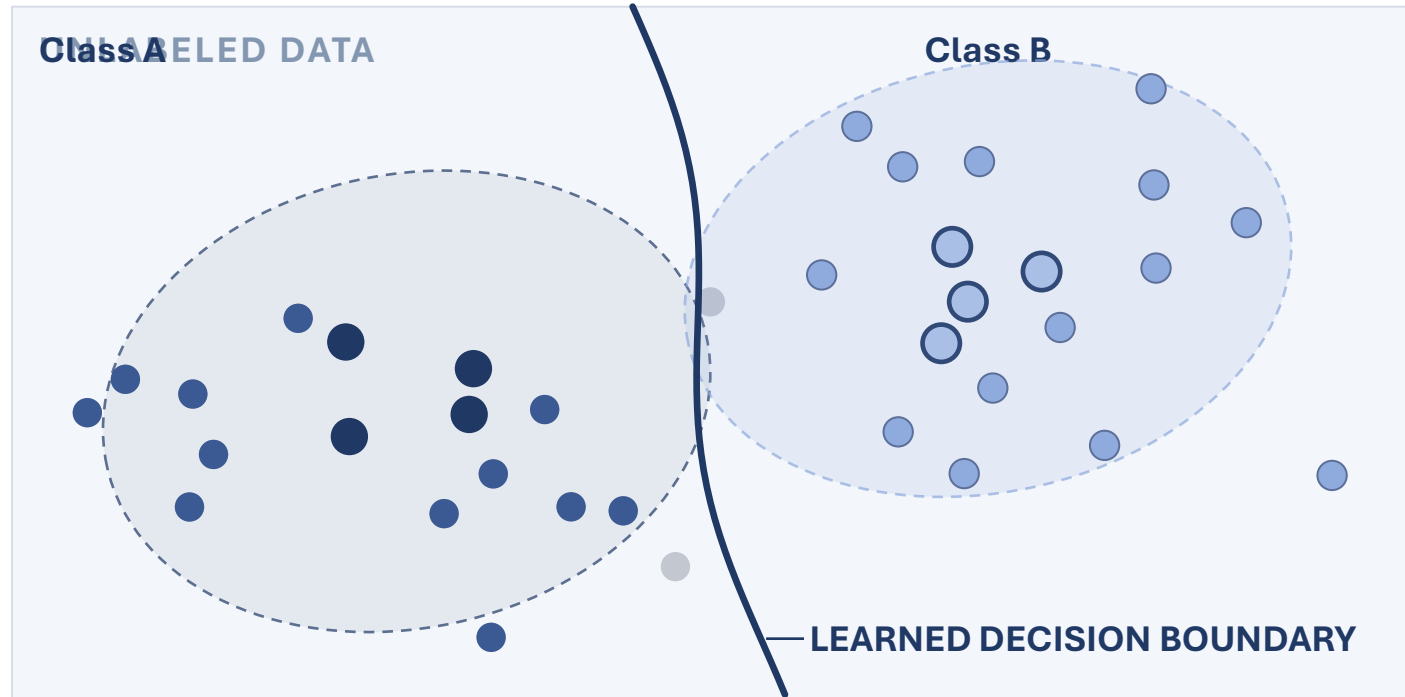
Discover patterns in unlabeled data

Also: Semi-Supervised Learning
Reinforcement Learning



SEMI-SUPERVISED LEARNING

Learn from a few labeled examples and many unlabeled examples.



●● FEW LABELED SAMPLES

● MANY UNLABELED SAMPLES

USE BOTH

Labeled + Unlabeled

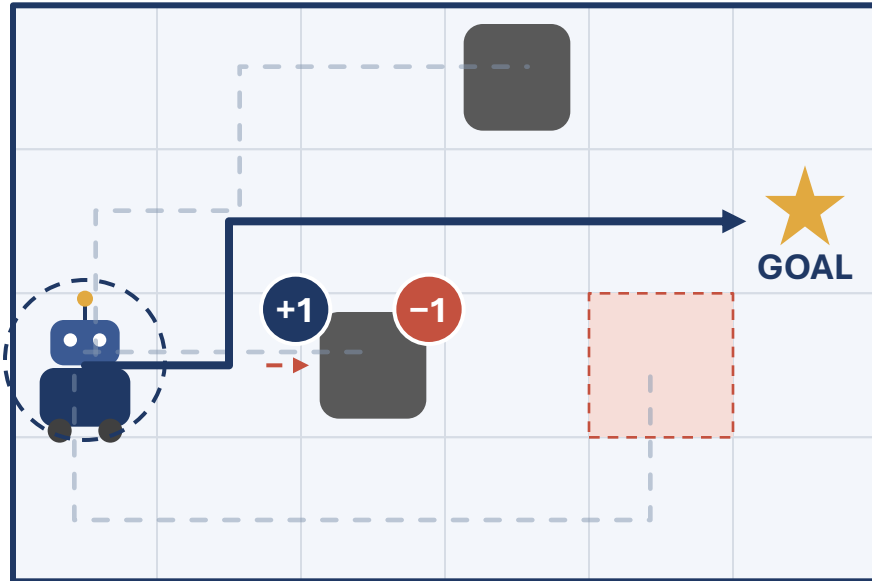
SUPERVISED
Labeled Data

UNSUPERVISED
Unlabeled Data

SEMI-SUPERVISED
Labeled + Unlabeled

REINFORCEMENT LEARNING

Learn by interacting with an environment and receiving feedback.



REWARD Did the agent reach the goal?

Repeating the process until the agent reaches the goal



Goal: learn a strategy that maximizes long-term reward.

REINFORCEMENT LEARNING
= LEARN THROUGH ACTIONS + FEEDBACK

VEVOX! CHALLENGE

Machine Learning Types

Let's see what you remember!



1. Scan



2. Join



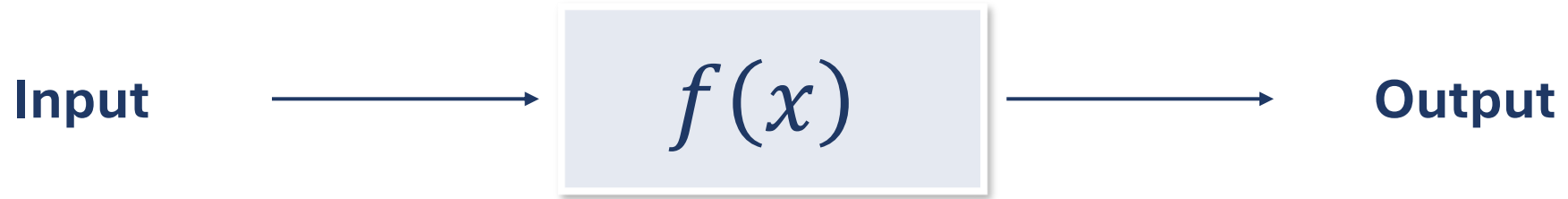
3. Compete

Join at:
vevox.app

ID:
423-890-106



Machine Learning in a Nutshell



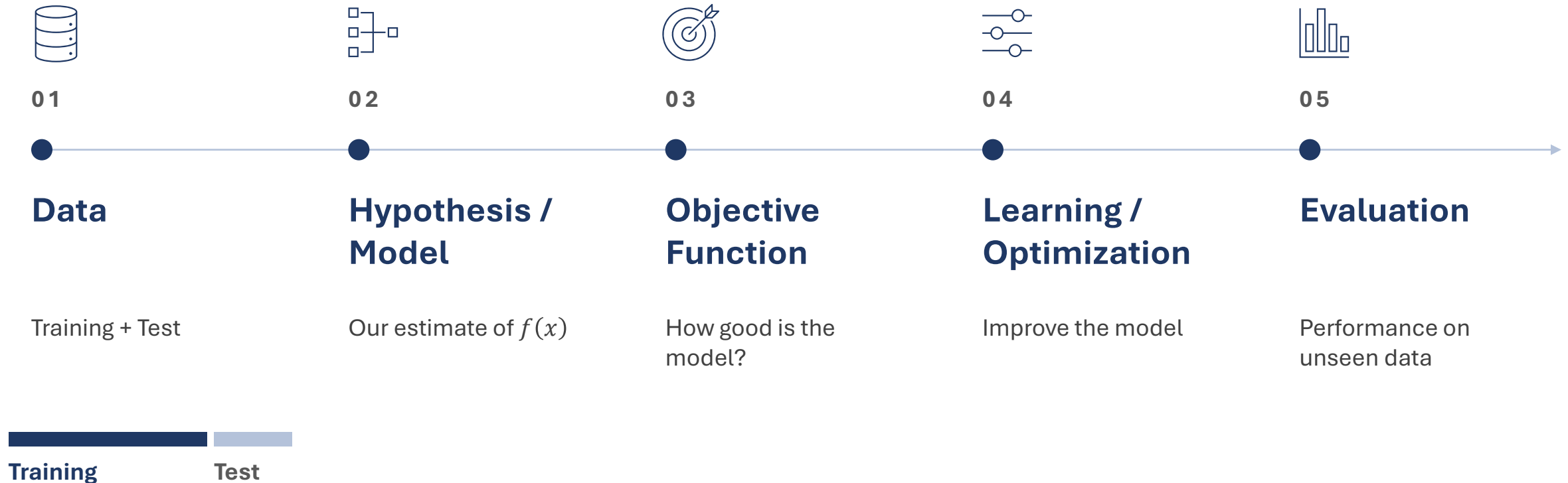
We assume an unobserved function in form of

$$y = f(x) + \varepsilon$$

where ε captures measurement errors and other discrepancies.

While the function is unobserved, it can be guessed

What Do We Need to Train a Machine Learning Model?



Supervised Learning

Learn from labeled examples

Each input comes with a known output. The model learns to predict the output for new inputs.

TWO COMMON TASKS

Regression

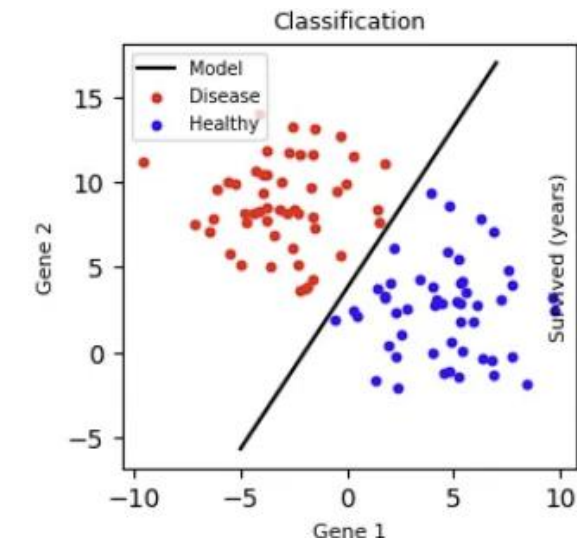
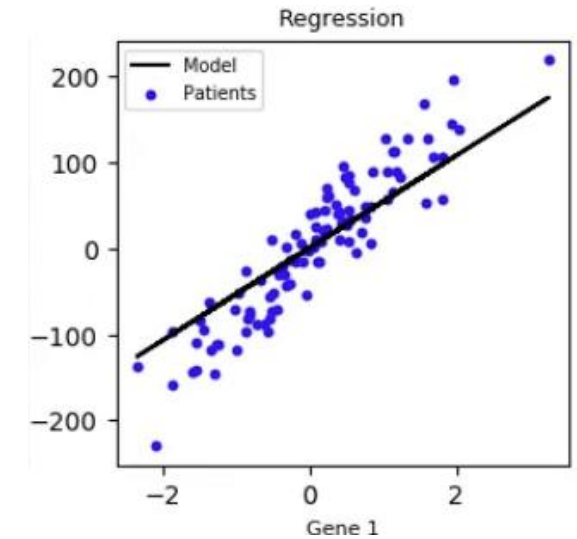
Predict a continuous value

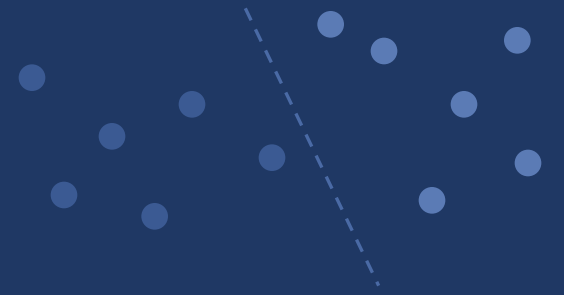
Output → **Number**

Classification

Predict a class / category

Output → **Category**





CLASSICAL MACHINE LEARNING

From learning problems to learning algorithms

How can we learn the relationship between inputs and outputs?

01

Linear Regression

02

Logistic Regression

03

SVM / SVR

04

**Decision Trees &
Random Forests**

Linear Regression

Model

$$\hat{y}_i = \beta_0 + \beta_1 x_i$$

Predict a continuous output with a straight line.

β_0 intercept β_1 slope

Prediction Error

$$e_i = y_i - \hat{y}_i$$

Actual – Predicted

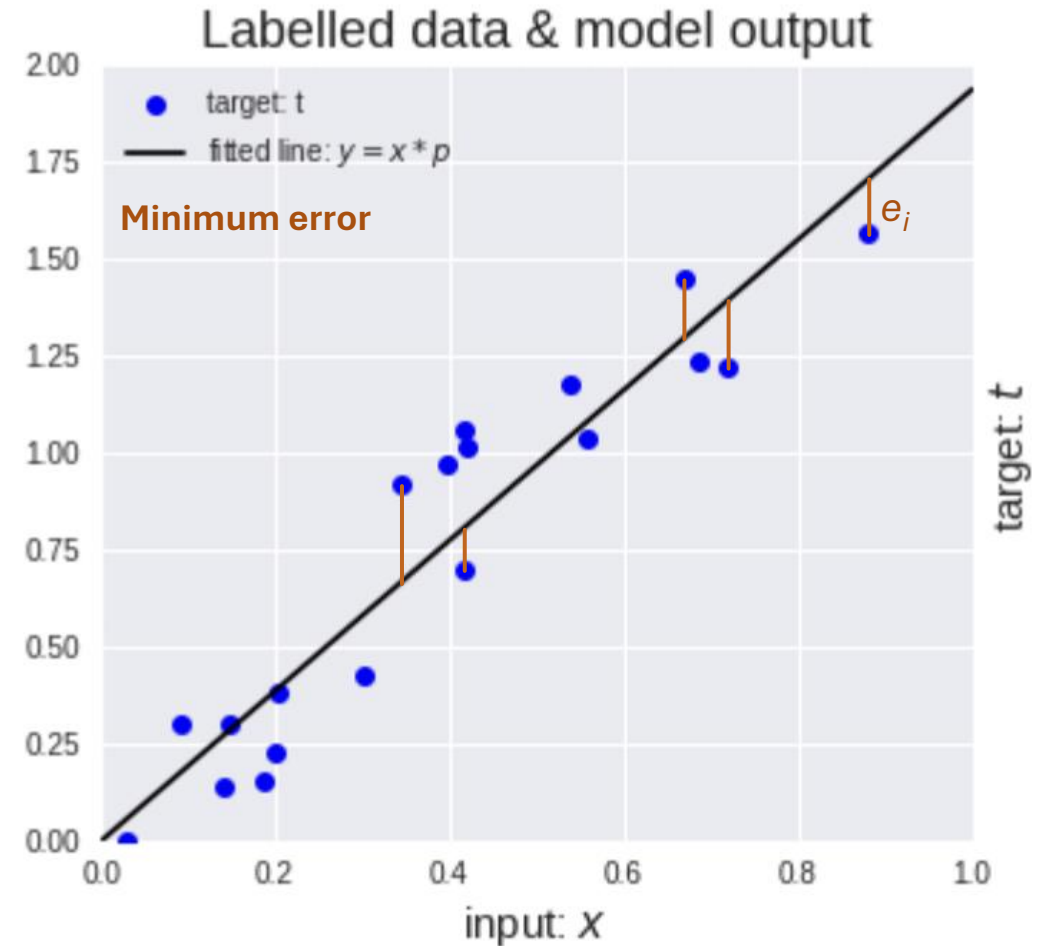
Objective

Find the line that minimizes prediction error.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{1}{n} \sum_{i=1}^n e_i^2$$

Smaller MSE → better fit

Training = finding β_0 and β_1 that **minimize the loss**



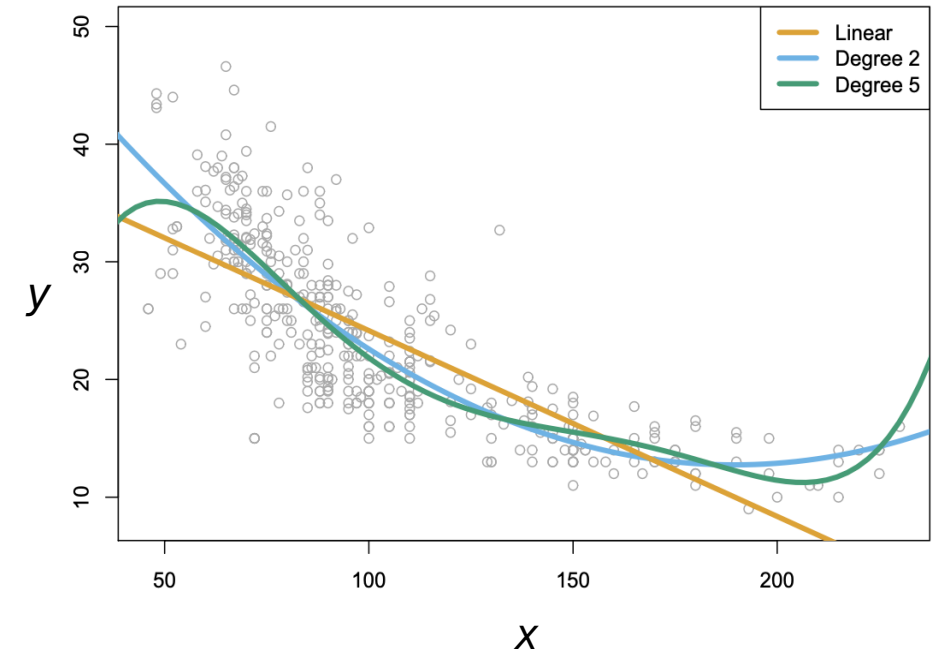
Polynomial Regression

What if a straight line is not enough?

$$\hat{y} = \beta_0 + \beta_1x + \beta_2x^2 + \cdots + \beta_nx^n$$

Higher degree $\rightarrow$ more flexible curve

Same regression idea, a more flexible model.



Too flexible? $\rightarrow$ Risk of overfitting

Logistic Regression

$$z = \beta_0 + \beta_1 x$$

Start with a linear combination

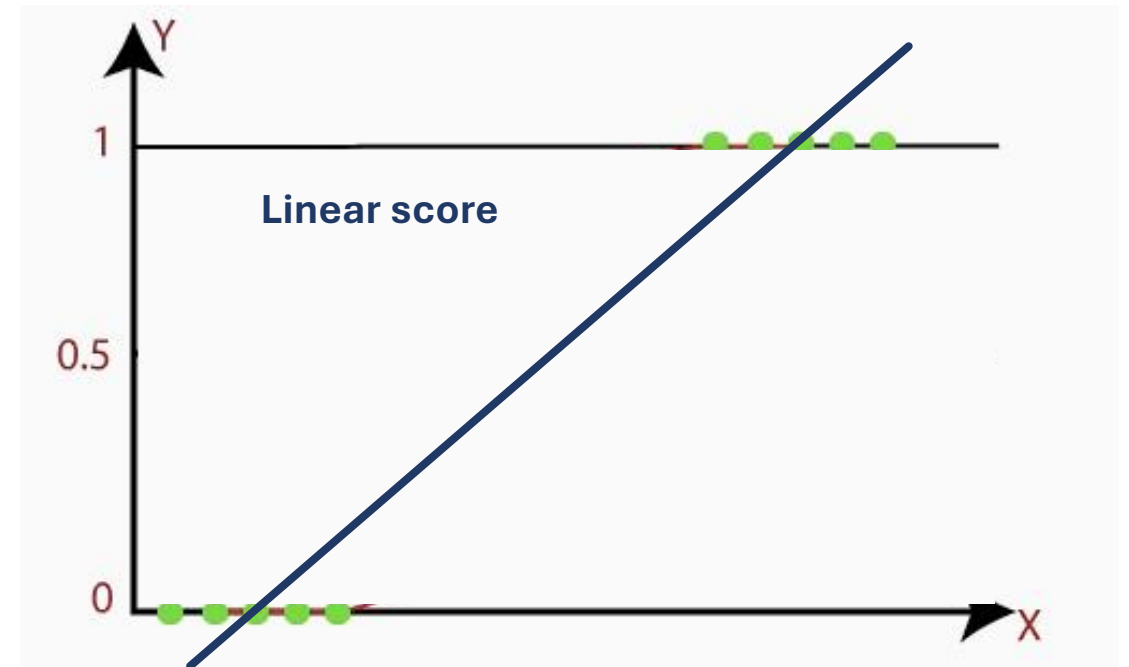
$$\hat{p} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

Map the score between 0 and 1

$$\hat{p} \geq 0.5 \rightarrow \text{Fault}$$

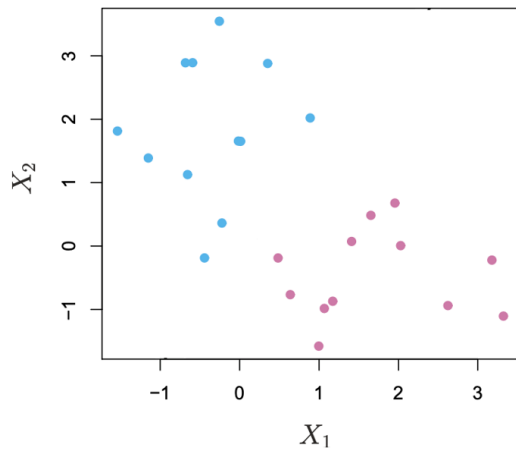
$$\hat{p} < 0.5 \rightarrow \text{Healthy}$$

Linear score $\rightarrow$ Probability $\rightarrow$ Class

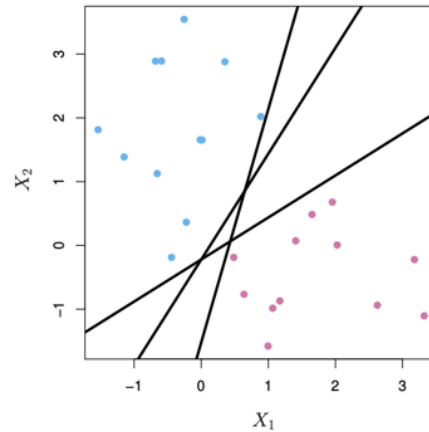


Support Vector Machine (SVM)

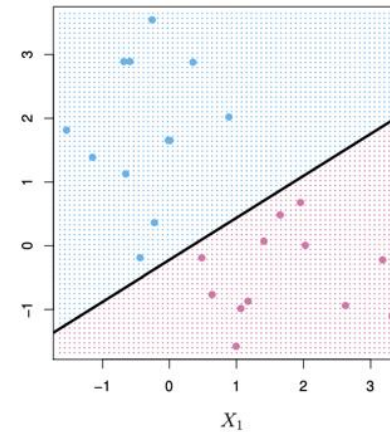
Find the boundary with the largest margin between classes.



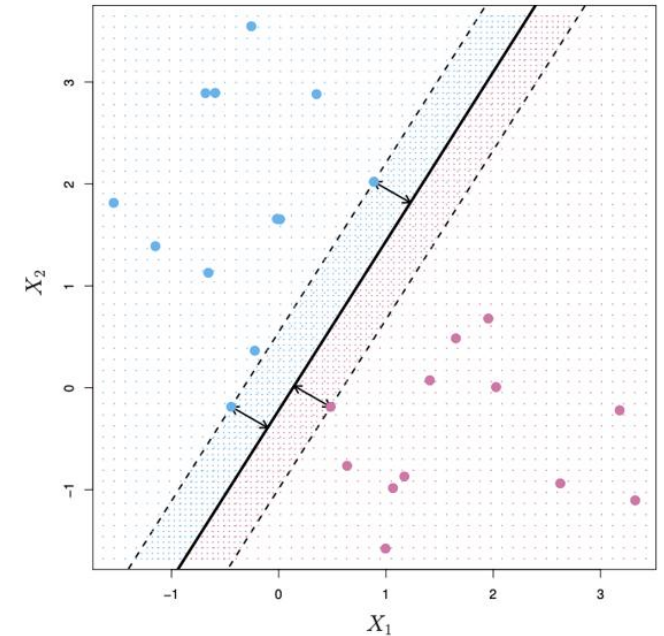
Two classes



Many boundaries can separate them



Choose a separating boundary



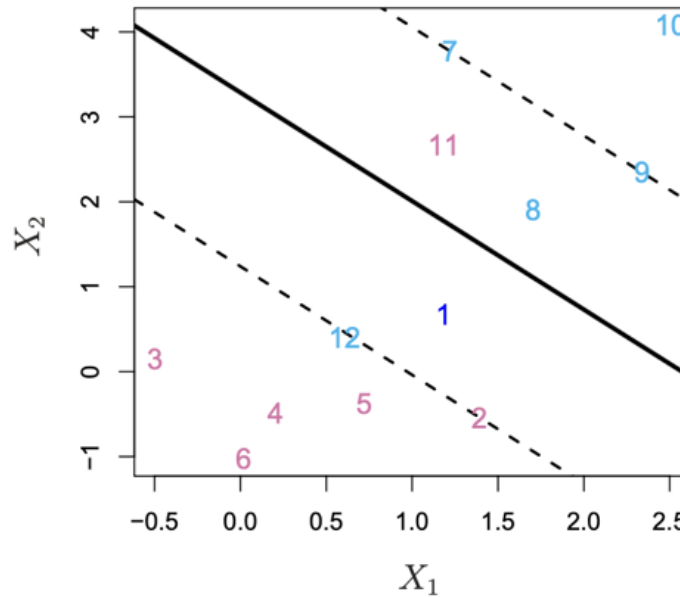
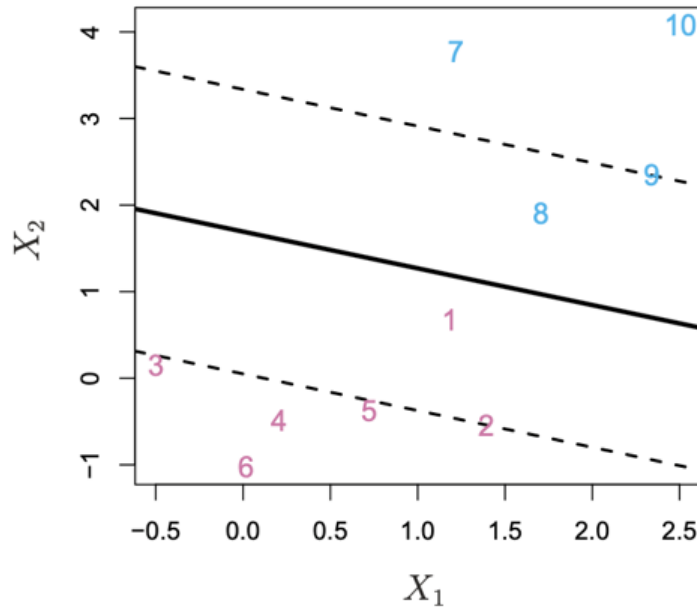
— Decision Boundary
Separates the classes

- - - Margin
Distance around the boundary

○ Support Vectors
Points closest to the margin

Maximize the margin → more robust separation

What if the data are not perfectly separable?



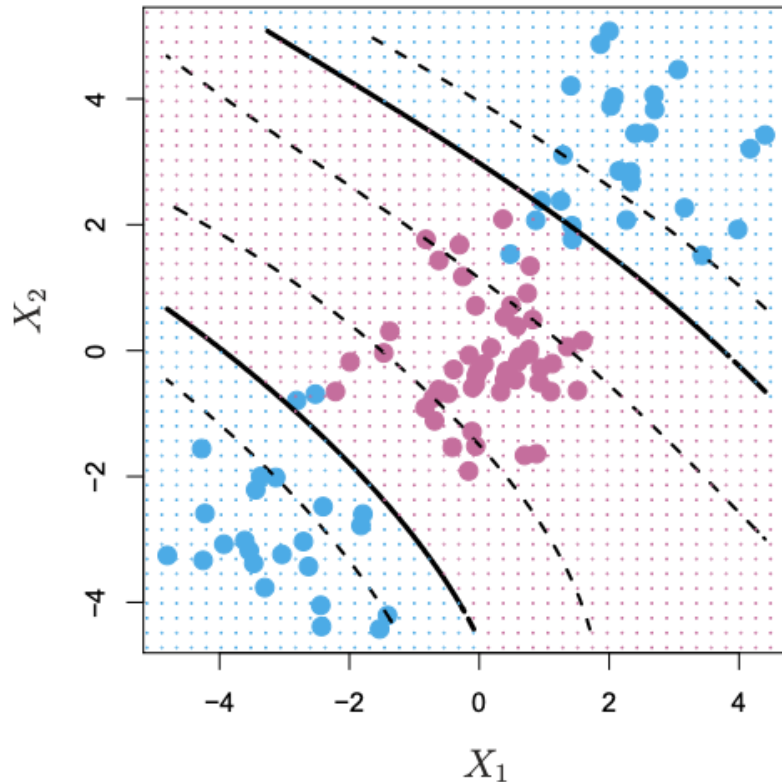
Soft-Margin SVM

- Allows some **margin violations**
- Balances **wide margin** vs. **classification error**
- Controlled by **C**

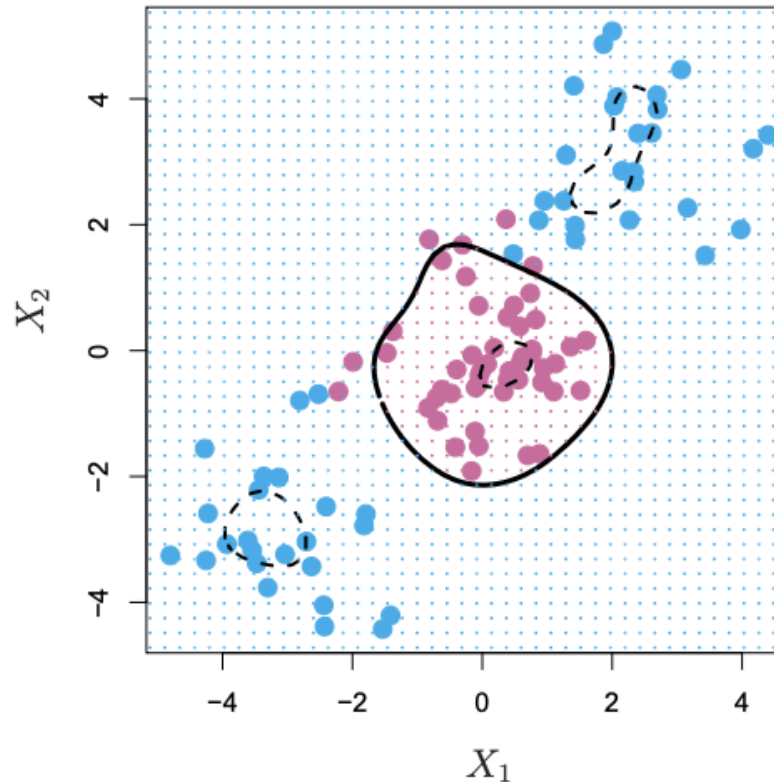
Some violations can improve generalization.

What if a straight boundary is not enough?

Polynomial / Cubic Kernel



RBF / Radial Kernel



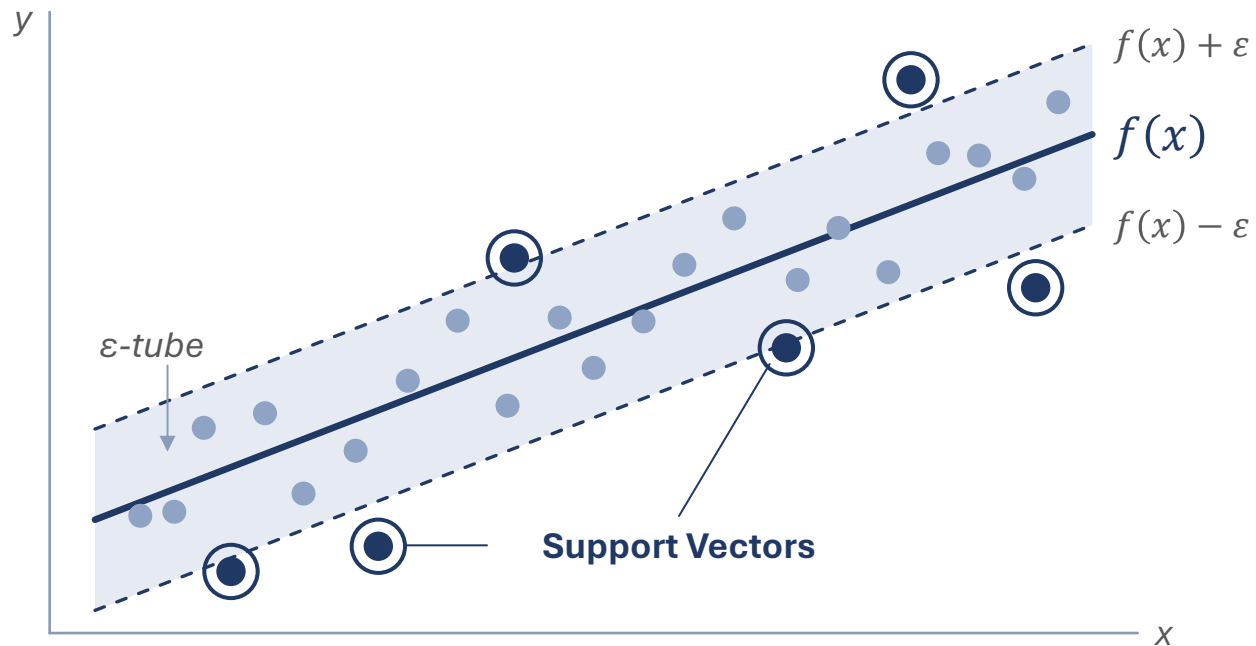
Kernels allow SVM to learn nonlinear decision boundaries.

Different kernels → different boundary shapes

Choose the kernel that best matches the structure of the data.

Support Vector Regression (SVR)

Fit a function while tolerating small errors.



$$|y_i - \hat{y}_i| \leq \varepsilon$$

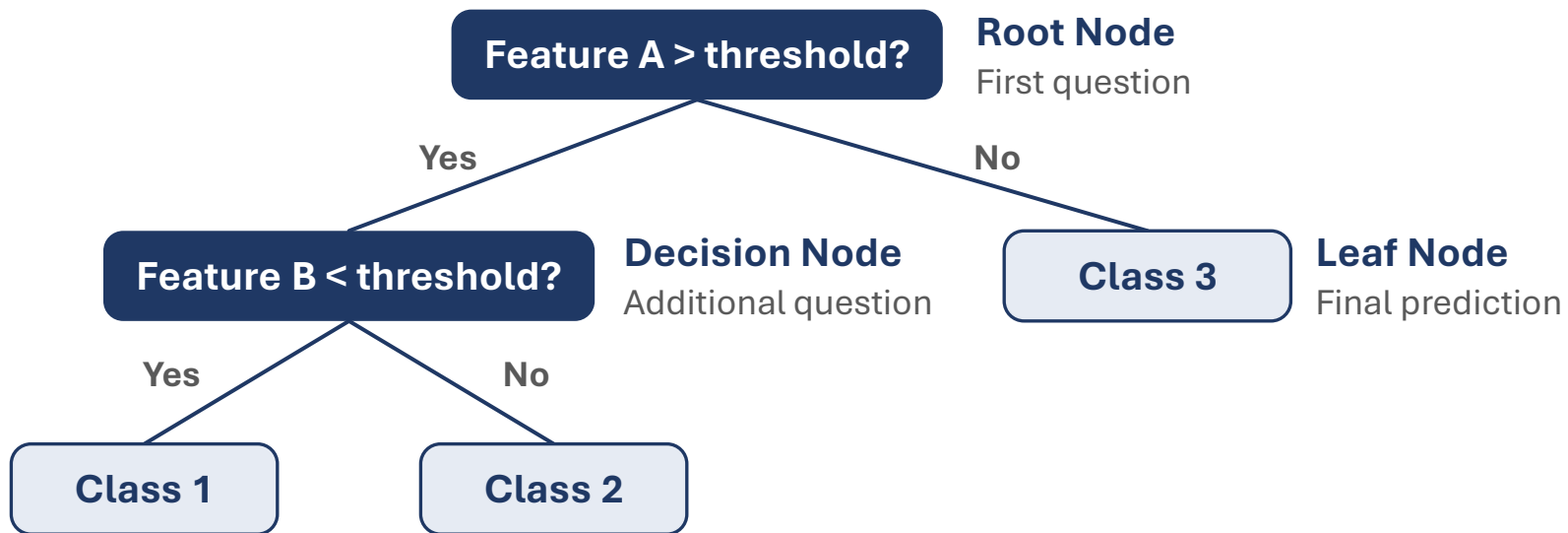
Errors inside the ε -tube are tolerated.

SVM → separate classes with a margin

SVR → fit values within an error margin

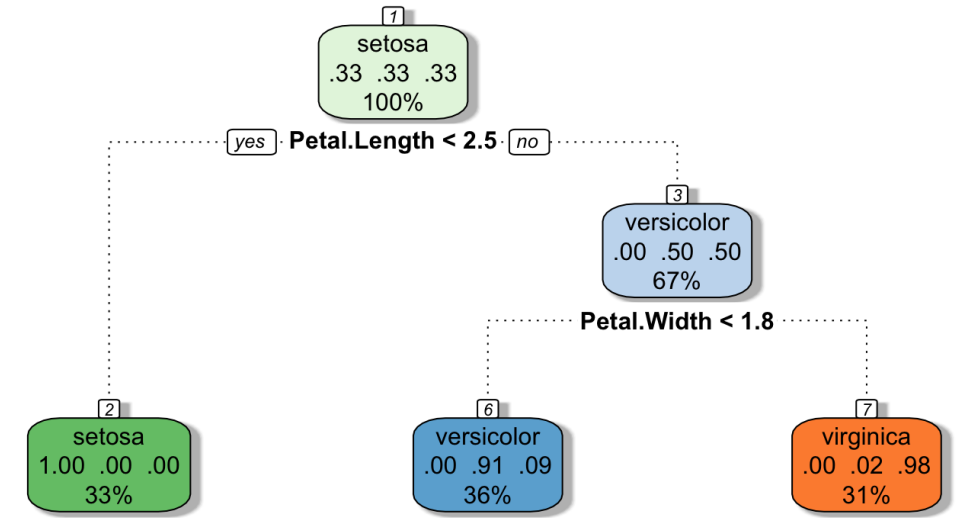
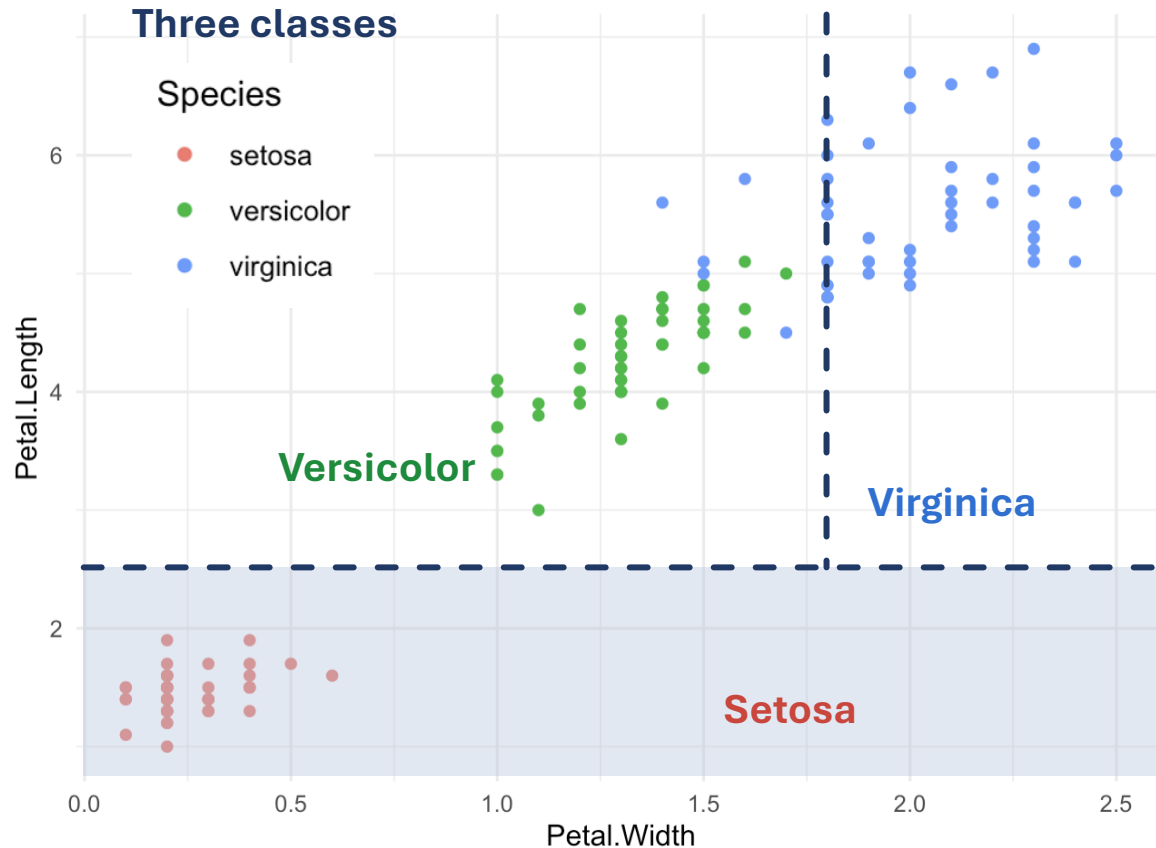
Decision Trees

A model that makes predictions by asking a sequence of questions.



Question → Split → Prediction

Decision Tree Example: Iris Dataset



Root Node

First question

Decision Node

Additional split

Leaf Node

Final prediction

Each question splits the data into smaller regions.

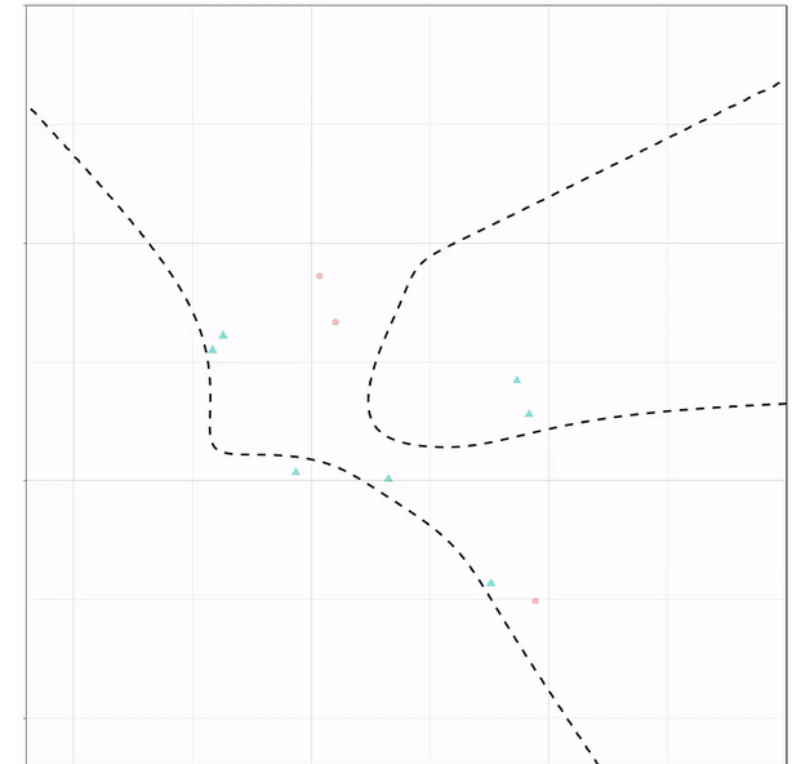
Random Forest

Combine many Decision Trees to make one prediction.



Each tree learns from a different sample and subset of features.

Many different trees → more robust prediction



Many trees → combined decision boundary

VEVOX! CHALLENGE

CLASSICAL MACHINE LEARNING

Let's see what you remember!



1. Scan



2. Join



3. Compete

Recognize the algorithm · Match the visual · Identify the task

Join at:
vevox.app

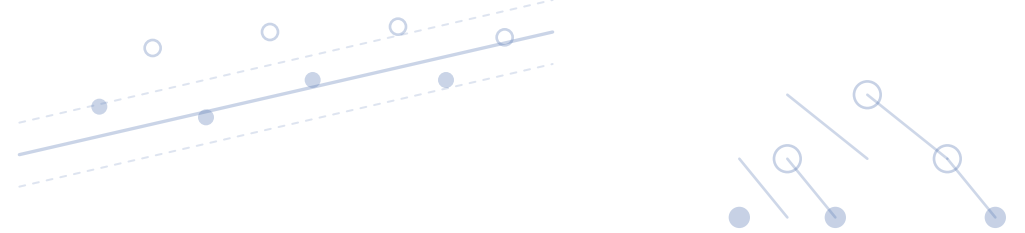
ID:
449-740-053



HANDS-ON ACTIVITY

Classical ML in Colab

~20 min



Train → Predict → Compare

Run the models you just learned and compare their behavior.

- 1 Load the Data
- 2 Train the Models
- 3 Run Predictions
- 4 Compare Results

COLAB NOTEBOOK

[Open Colab Notebook →](#)

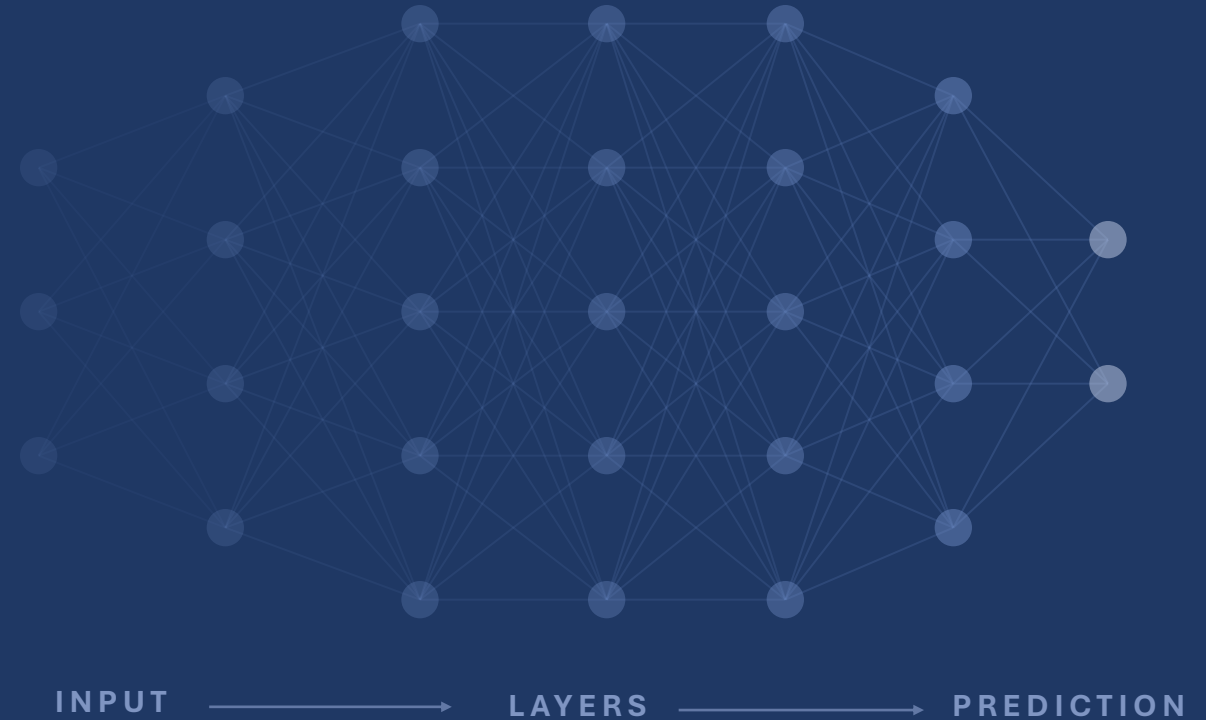
Click to open in your browser



DEEP LEARNING

Learning complex patterns through
neural networks

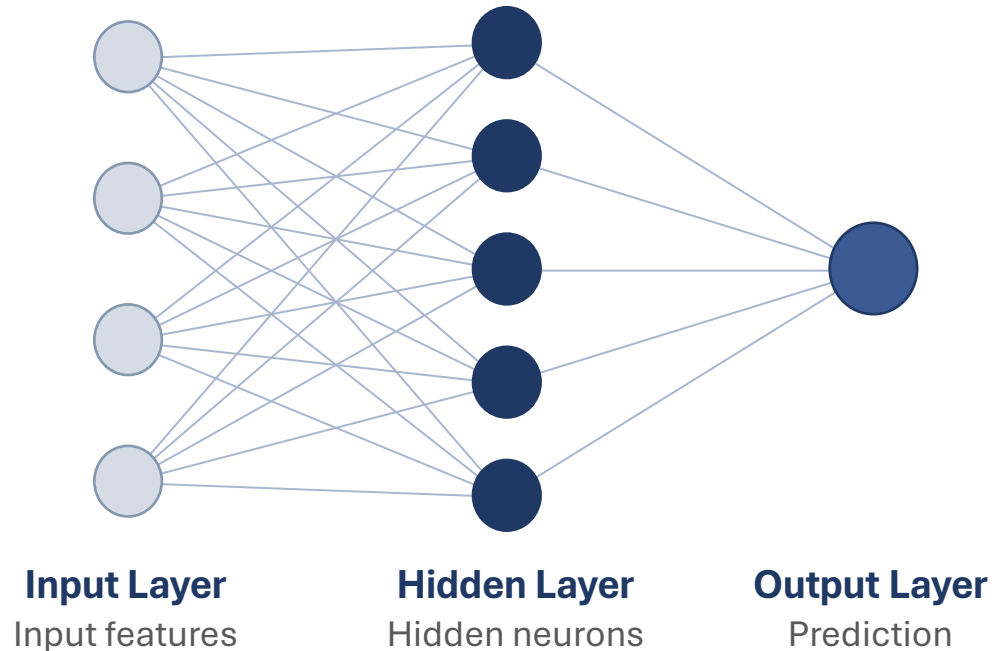
What if the model could learn useful features
directly from the data?



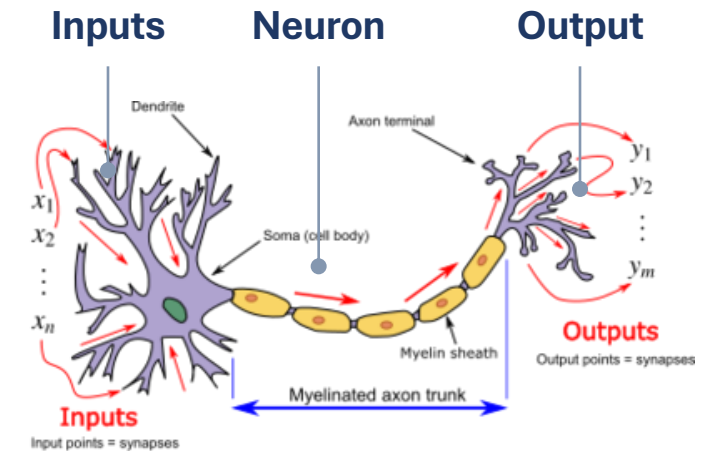
Neural Networks · CNNs · RNNs (LSTM / GRU)

From Biological Neurons to Neural Networks

ARTIFICIAL NEURAL NETWORK



BIOLOGICAL NEURON

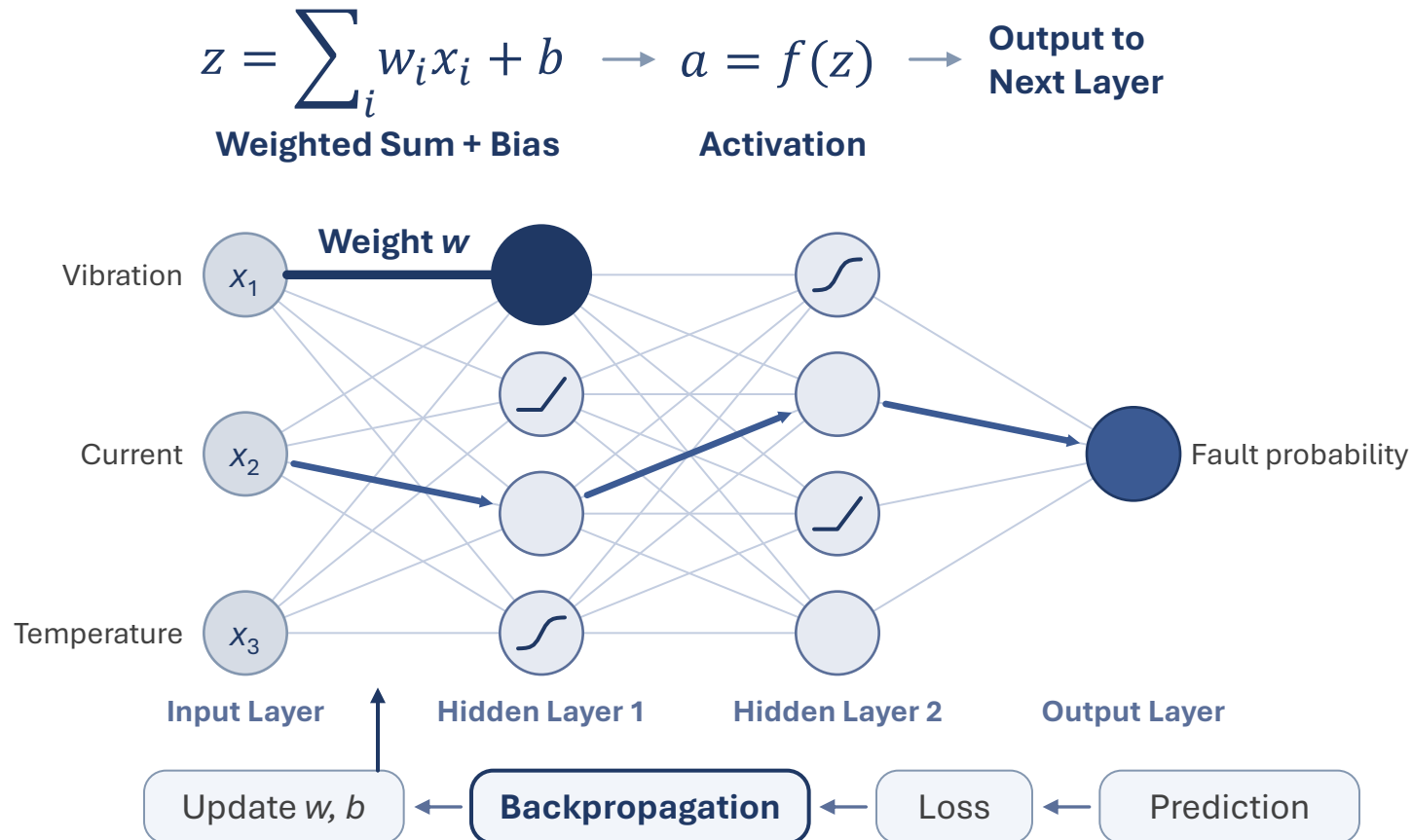


Neural networks use a simplified mathematical abstraction of biological information flow.

Can we represent this behavior mathematically?

How Does a Neural Network Work?

FORWARD PASS Inputs → Hidden Layers → Prediction



ACTIVATION FUNCTIONS

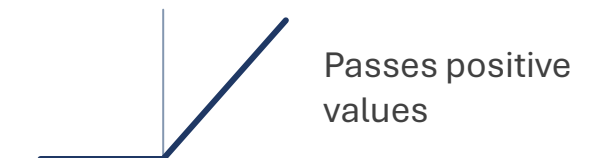
Sigmoid

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$



ReLU

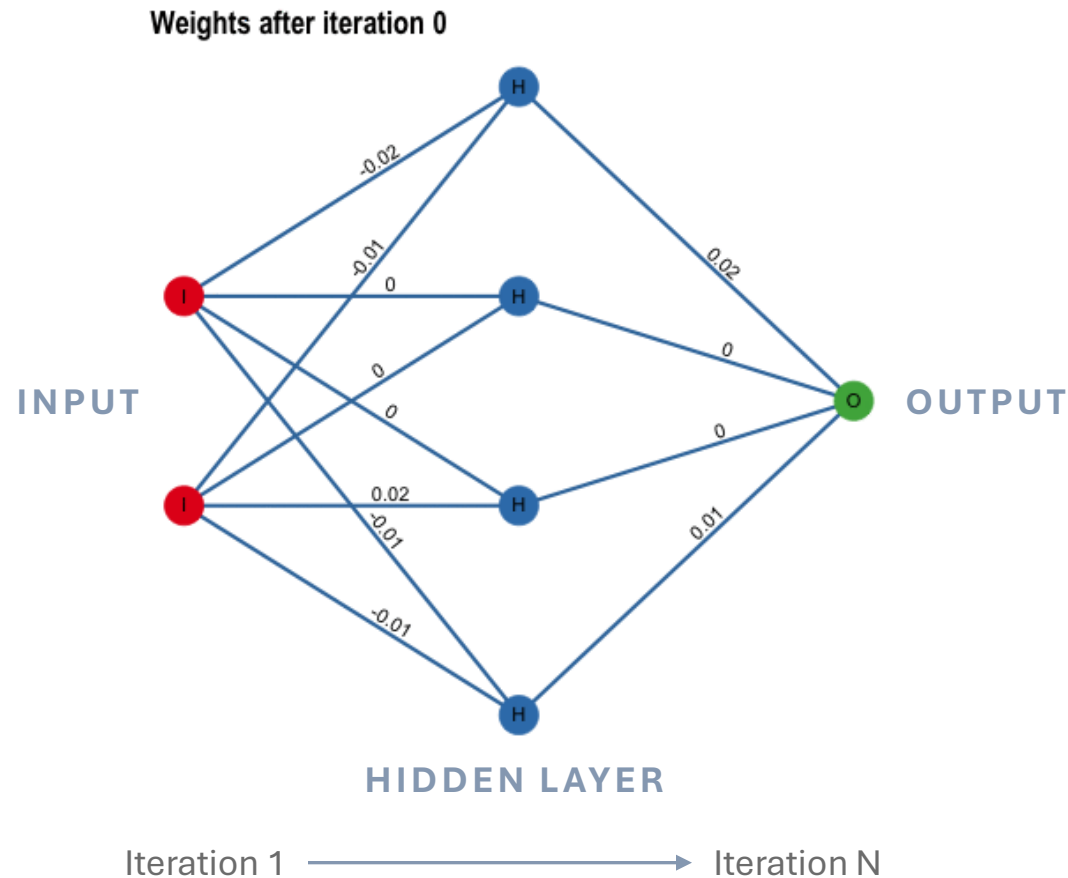
$$\text{ReLU}(z) = \max(0, z)$$



Forward pass makes the prediction.
Backpropagation improves the parameters.

HOW DOES THE NETWORK LEARN?

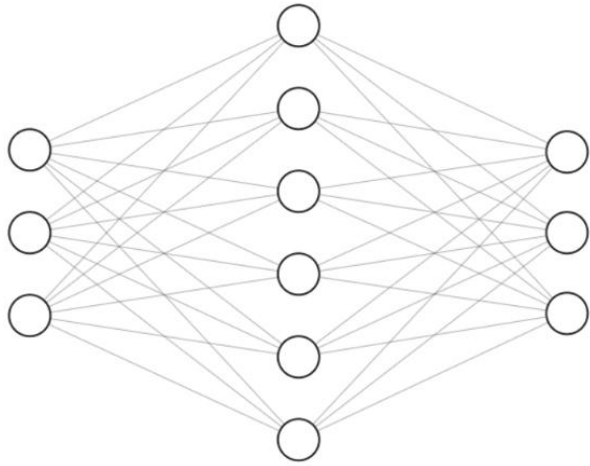
Weights and biases update during training



Training adjusts the **weights and biases** to **reduce the error**.

Shallow vs. Deep Neural Networks

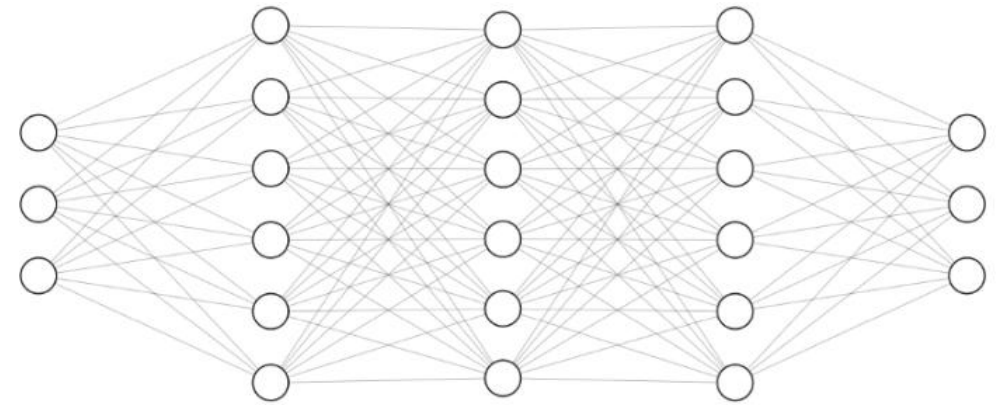
Shallow Neural Network



One hidden layer

Input → Hidden Layer → Output

Deep Neural Network



Multiple hidden layers

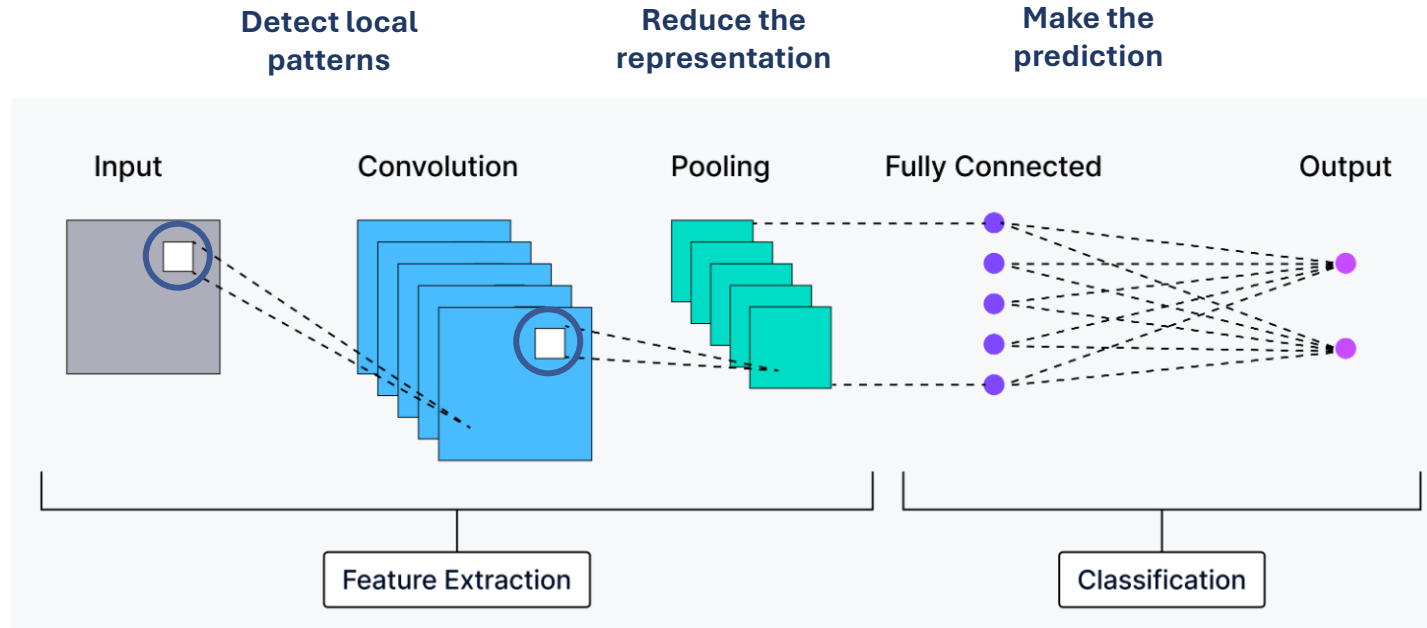
Input → Hidden Layers → Output

“Deep” refers to the number of hidden layers, not simply the number of neurons.

More hidden layers allow the network to learn increasingly complex representations.

Convolutional Neural Networks (CNNs)

CNNs learn local patterns automatically.



○ **Filter / Kernel**
Scans local regions for patterns

Feature maps
Show where patterns appear

Convolution learns useful local features before the network makes a prediction.

CNNs are not only for images

1D CNN



Vibration · Current ·
Time-Series Signals

2D CNN



Images · Spectrograms

EXPLORE A CNN

Interactive exploration

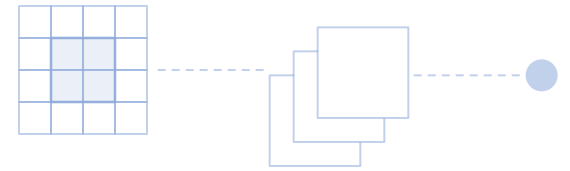
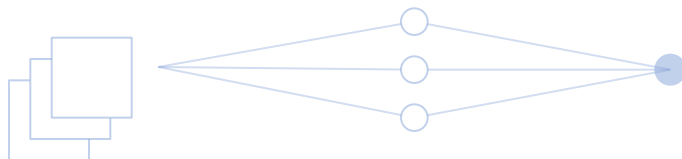
~5 min

Search “CNN Explainer”

Open the first result — [CNN Explainer](#)

Explore how an image moves through the network.

- 1 Convolution**
Watch how the filters move across the input.
- 2 Feature Maps**
See what different filters detect.
- 3 Prediction**
Follow the features through to the final output.

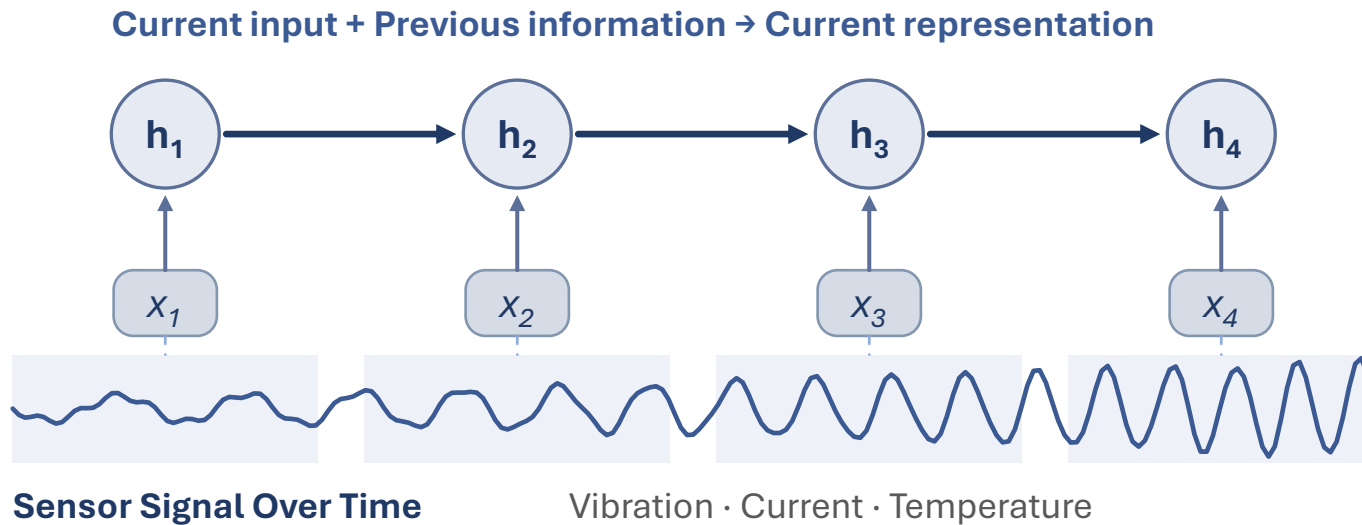


CHALLENGE

What changes as you move deeper into the network?

Recurrent Neural Networks (RNNs)

What if the order of the data matters?



But how long can an RNN remember?

Standard RNNs can struggle with long-term dependencies.

RNNs use previous information to understand the current point in a sequence.

RNNs keep information from previous time steps.

CNN VS. RNN

CNN Local patterns

RNN Sequential patterns



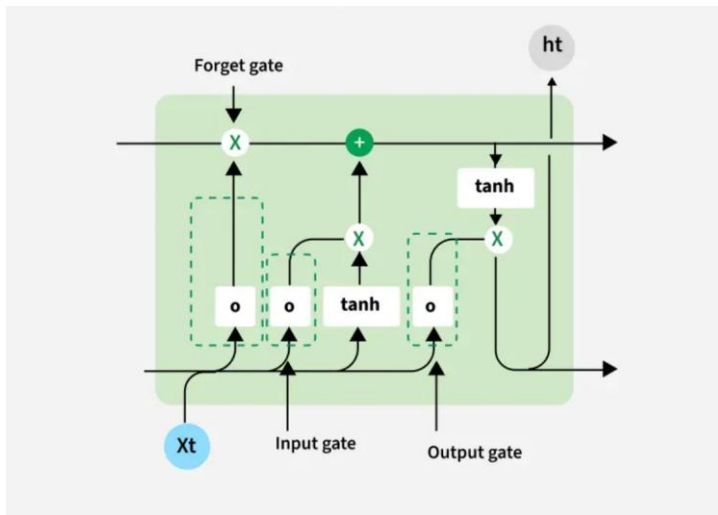
Better control of memory

LSTM & GRU

How can we better control what an RNN remembers?

Gates control what information is kept, updated, or forgotten.

LSTM



Forget • Input • Output

More control over memory

LSTM

3 main gates

Separate cell state + hidden state

GRU

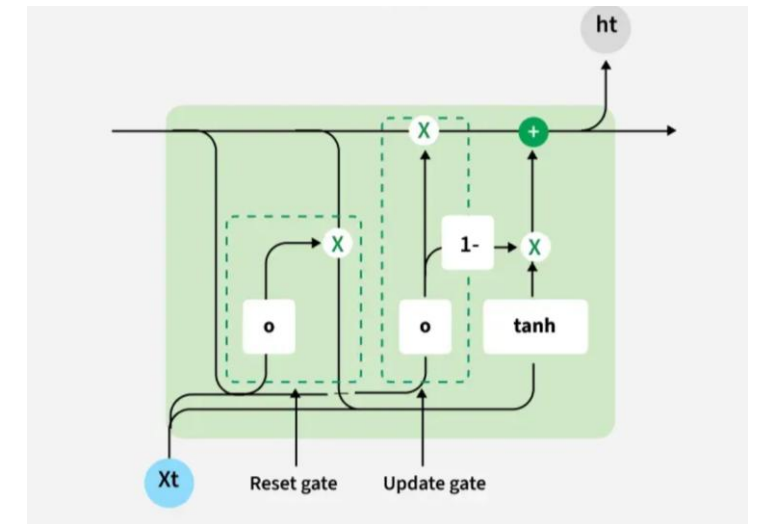
2 main gates

More compact structure

RNN → Basic sequence memory

LSTM / GRU → Gated memory

GRU



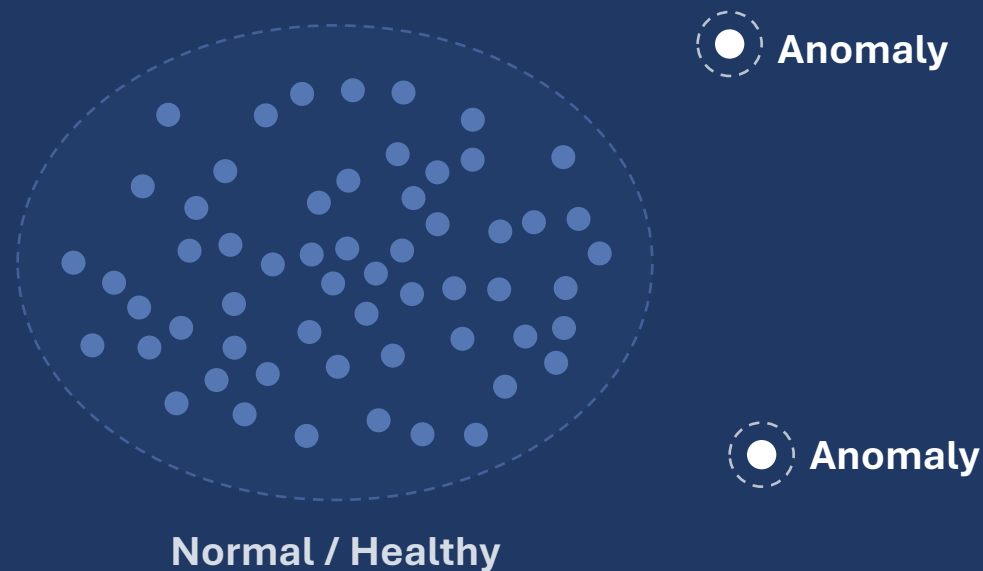
Reset • Update

Simpler gating structure

LSTM and GRU are designed to better handle long-term dependencies in sequential data.

ANOMALY DETECTION

What if we know what “normal” looks like... but have very little fault data?



Learn Normal → Detect Deviation → Flag Anomaly

The Challenge with Fault Data

Why not just train a classifier?

Data Scarcity

Healthy Data



Fault Data



Fault examples are difficult to collect.

Rare · Expensive · Unknown

Class Imbalance

Healthy — 98%

Fault — 2%



Most samples belong to the healthy class.

High accuracy ≠ Good fault detection



So how can we detect unusual behavior under these conditions?

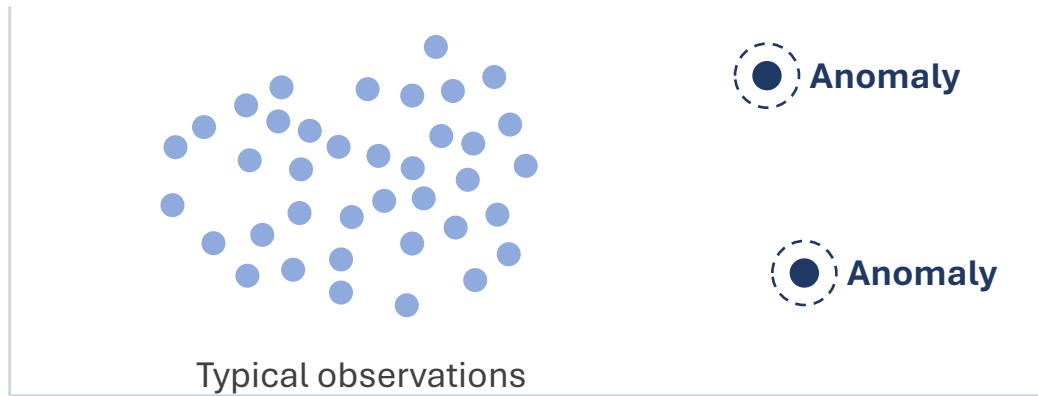
Key Challenges in Anomaly Detection

Limited abnormal data · Imbalanced datasets · Rare or unseen behaviors

Anomaly Detection — General Overview

An anomaly is an **observation or pattern** that differs significantly from expected behavior.

Feature Space



Time Series



Data → **Detection Method** → **Anomaly Score** → **Decision**

$s(x)$ → **Threshold** → **Normal / Anomalous**

x can be a sample, feature vector, or time window.

Anomaly ≠ Fault

An anomaly indicates unusual behavior; the physical cause still requires interpretation.

Distance · Density · Reconstruction Error · Learned Boundary

Common Anomaly Detection Methods

Different approaches to identifying unusual behavior

01

K-Means

02

**Gaussian Mixture
Models (GMM)**

03

**Autoencoders /
One-Class Learning**

Different methods. Different perspectives on what is unusual.

K-Means Clustering

Group similar observations around K centroids.

1

Choose K

$K = 4$ clusters

2

Initialize

Place K centroids

3

Assign (Assignment Step)

Each point $\rightarrow$ nearest centroid

4

Update (Update Step)

Centroid $\rightarrow$ mean of assigned points

Repeat 3 $\leftrightarrow$ 4

Repeat until the centroids stop changing

$$\arg \min_k \|x_i - \mu_k\|^2$$

Assign x_i to its nearest centroid μ_k

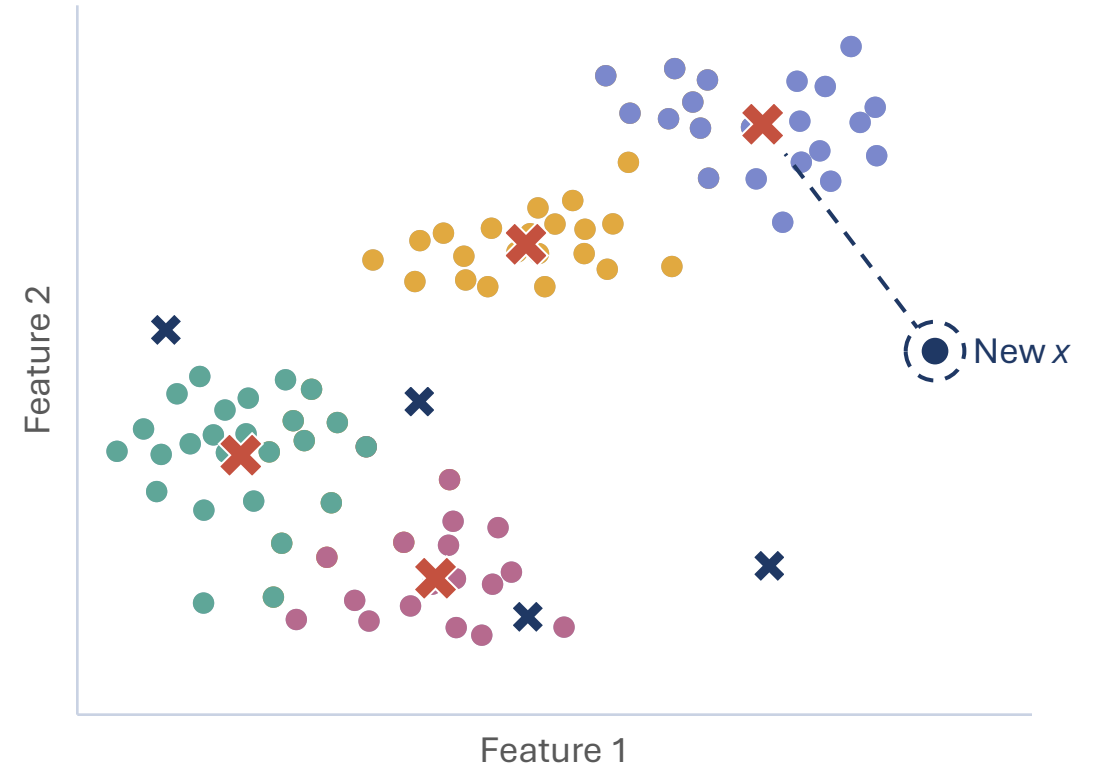
How can this help detect anomalies?

$$s(x) = \min_k \|x - \mu_k\|$$

Large distance $\rightarrow$ potentially unusual

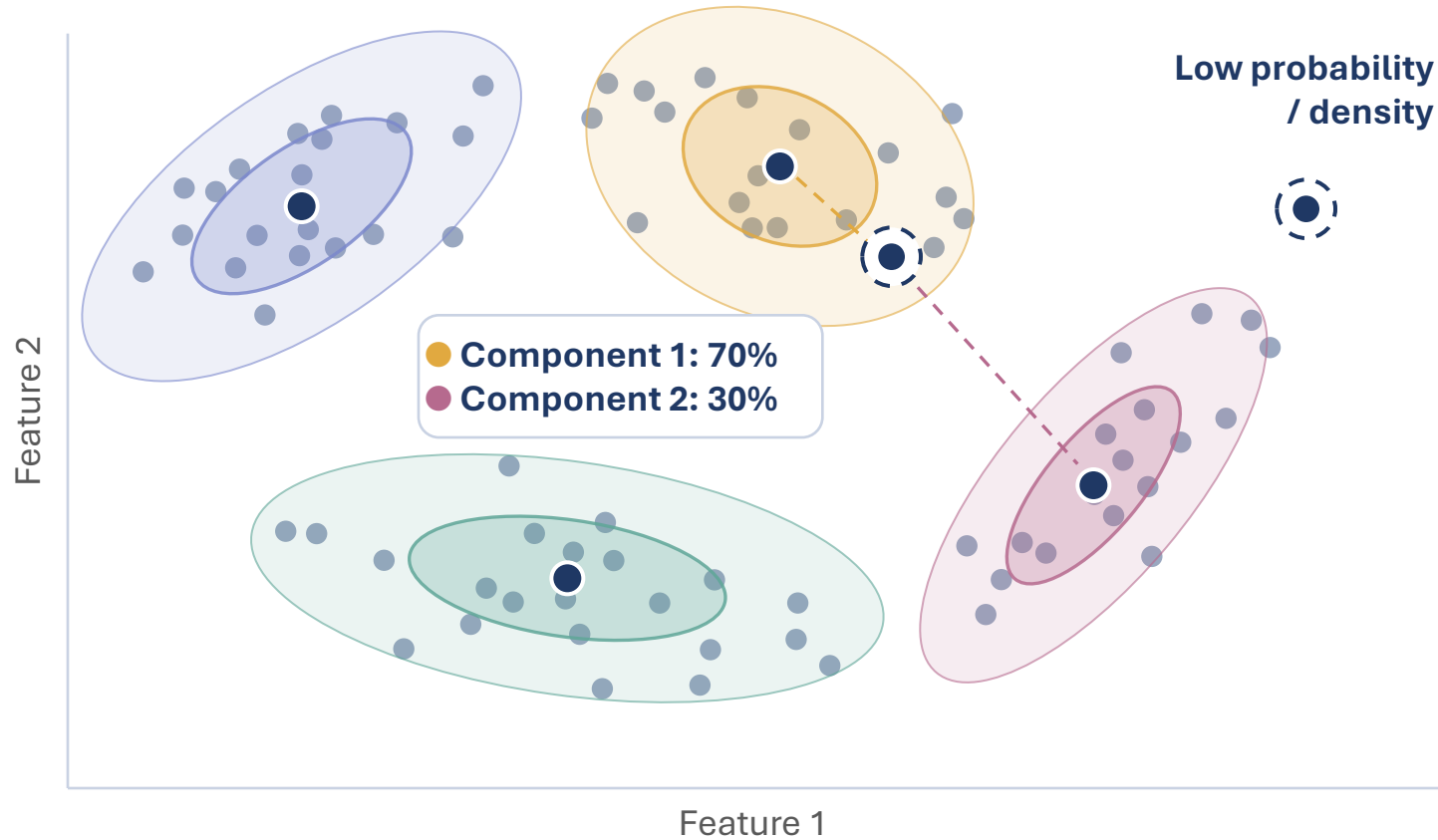
Converged

Centroids are no longer changing



Gaussian Mixture Models (GMM)

Model the data as a mixture of probability distributions.



Using GMM for Anomaly Detection

$p(x)$ small $\Rightarrow$ more unusual

$$s(x) = -\log p(x)$$

Higher score $\rightarrow$ more unusual

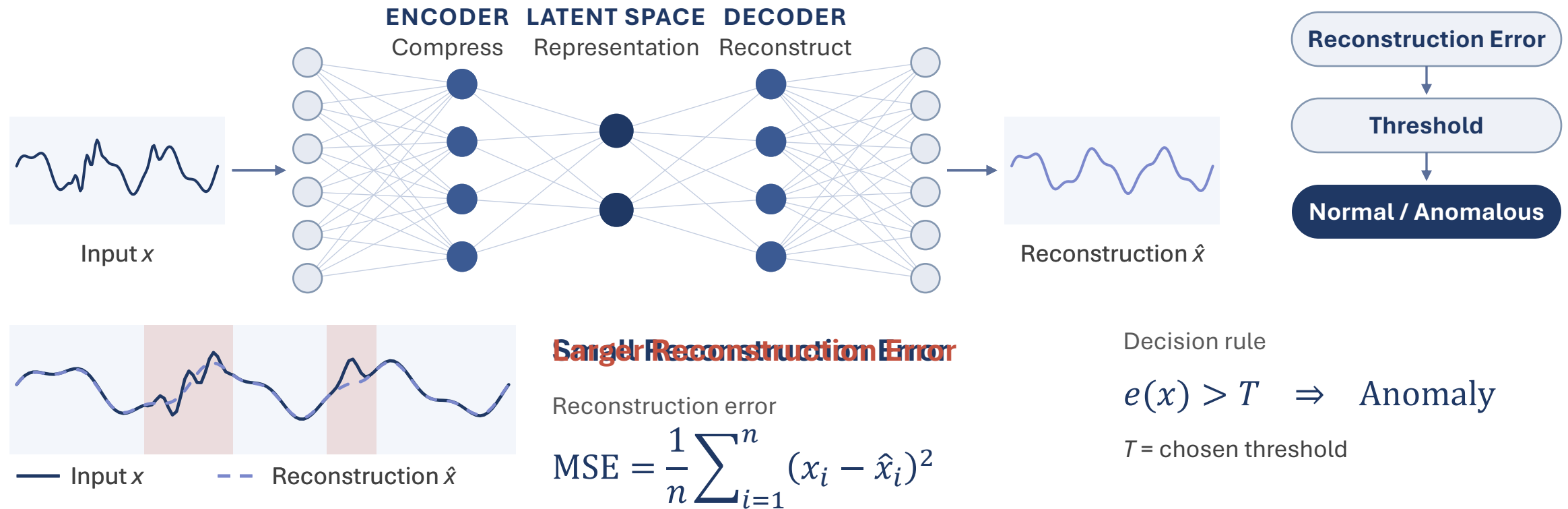
different probabilities.

K-Means · nearest centroid $\rightarrow$ **hard**

GMM · probability $\rightarrow$ **soft**

AUTOENCODERS FOR ANOMALY DETECTION

Can the network reconstruct its input?

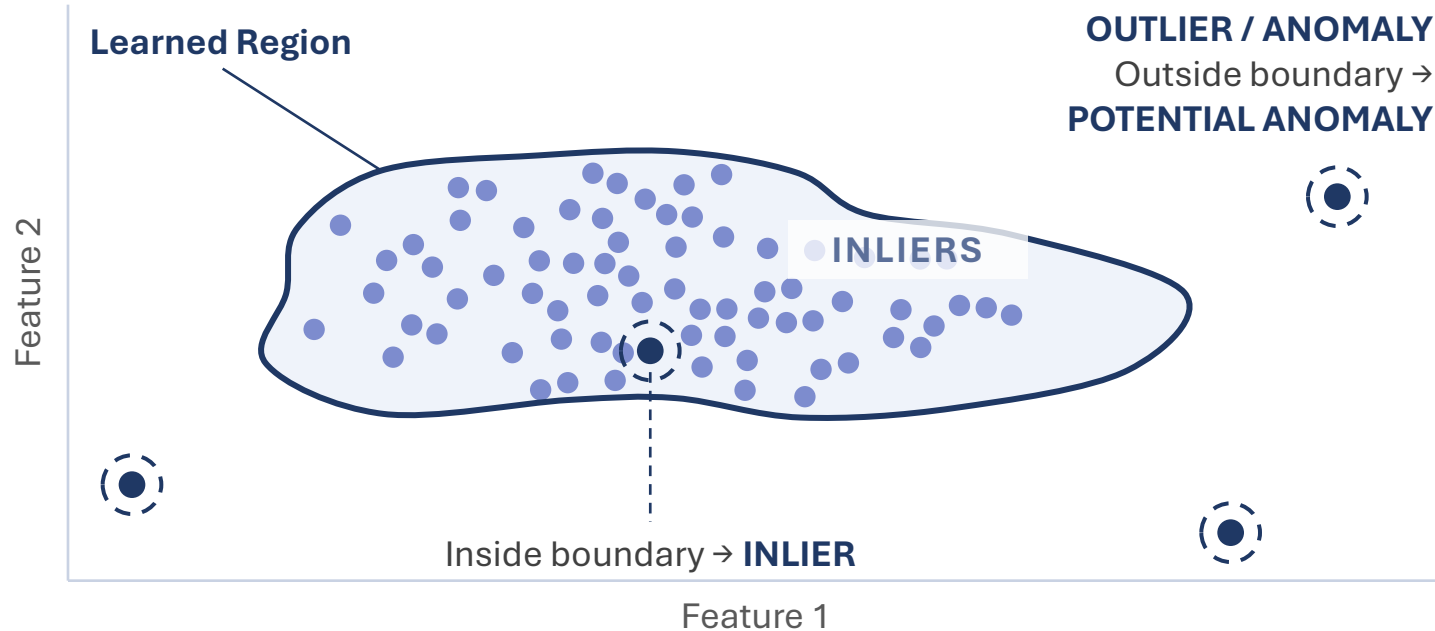


Reconstruction error can provide an anomaly score.

K-Means · distance **GMM** · probability / density **Autoencoder** · reconstruction error

ONE-CLASS LEARNING

What if we mainly have examples from one class?



What if the region occupied by the data of interest is not the data examples we have?

ONE-CLASS SVM

Standard SVM

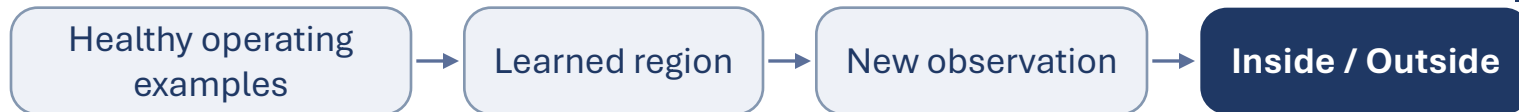
Separate Class A from Class B

One-Class SVM

Learn a boundary around the data of interest

One-Class Learning
→ Learned Boundary

In fault detection



Often useful when training data primarily represent one operating class.

K-Means
DISTANCE

GMM
DENSITY

Autoencoder
RECONSTRUCTION ERROR

**One-Class
BOUNDARY**



VEVOX! CHALLENGE

ANOMALY DETECTION

~10 min

Let's see what you remember!

Scan. Join. Compete.



1 — Scan



2 — Join

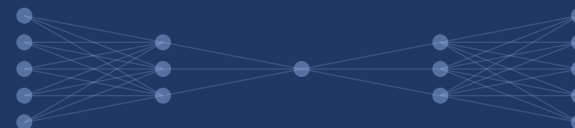


3 — Compete

Recognize the method · Interpret the anomaly · Match the concept

Join at:
vevox.app

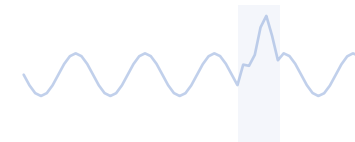
ID:
497-468-751



HANDS-ON ACTIVITY

Anomaly Detection in Colab

~30 MIN

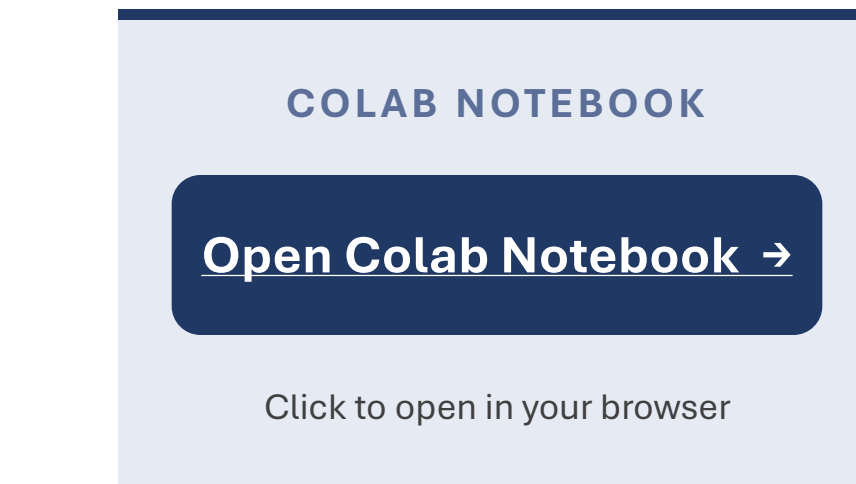


Explore → Detect → Compare

Explore how different methods identify unusual observations.

- 1 EXPLORE THE DATA**
Visualize typical and unusual observations
- 2 RUN THE METHODS**
Apply the prepared anomaly-detection models
- 3 DETECT ANOMALIES**
Inspect the anomaly scores and predictions
- 4 COMPARE**
See how methods respond to the same data

K-Means • GMM • Autoencoder • One-Class



Same data. Different definitions of “unusual.”



PRACTICAL SETUP

EDGE IMPULSE INSTALLATION GUIDE

Let's get everything set up.

